t-Linear Tensor Subspace Learning for Robust Feature Extraction of Hyperspectral Images

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***Abstract*— Subspace learning has been widely applied for feature extraction of hyperspectral images (HSIs) and achieved great success. However, the current methods still leave two problems that need to be further investigated. First, those methods mainly focus on finding one or multiple projection matrices for mapping the high-dimensional data into a low-dimensional subspace, which can only capture the information from each direction of high-order hyperspectral data separately. Second, the performance of feature extraction is barely satisfactory when the hyperspectral data is severely corrupted by noise. To address these issues, this article presents a t-linear tensor subspace learning (tLTSL) model for robust feature extraction of HSIs based on t-product projection. In the model, t-product projection is a new defined tensor transformation way similar to linear transformation in vector space, which can maximally capture the intrinsic structure of tensor data. The integrated tensor low-rank and sparse decomposition can effectively remove the noise corruption and the learned t-product projection can directly transform the high-order hyperspectral data into a subspace with information from all modes comprehensively considered. Moreover, a proposition related to tensor rank is proofed for interpreting the meaning of the tLTSL model. Extensive experiments are conducted on two different kinds of noise (i.e., simulated and real noise) corrupted HSI data, which validate the effectiveness of tLTSL.**

***Index Terms*— Hyperspectral image (HSI), multilinear projection, robust feature extraction, t-product, tensor subspace learning.**

I. Introduction

IN THE past decades, with the rapid development of information acquisition technology, both the views and dimensions of acquired data are sharply increasing, resulting in that high-order data analysis becomes a hot topic in computer vision. Although the high-order data can provide more valuable information, it also greatly increases the difficulty of data analysis. Especially, the so-called "curse of dimensionality" has become a challenging problem to address. How to extract the most representable features from the high-order data without useful information loss is also a prominent problem in hyperspectral image (HSI) processing [1], [2].

Up to now, numerous feature extraction methods have been proposed and successfully applied for HSI processing [3]. Among those methods, (linear) subspace learning is one of the most popular techniques for feature extraction of HSI. Generally, subspace learning can be simply divided into unsupervised and supervised methods. The typical unsupervised methods includes principle component analysis (PCA) [4], locality preserving projection (LPP) [5], etc. Linear discriminative analysis (LDA) and its extensions are the most popular supervised subspace learning methods in practice [6], [7], [8], [9]. Recently, due to the advantage of exploiting nonlinear structure, manifold learning has been another widely applied feature extraction technique [10], [11], [12]. The sparse representation (SR) and low-rank representation (LRR)-based methods are also powerful tools for HSI processing [13]. Sparse graph-based discriminant analysis (SGDA) was developed for feature extraction of HSIs in [14]. Li et al. [15] proposed sparse and low-rank graph-based discriminant analysis (SLGDA) to simultaneously exploit the locality and global structures of HSI data. Pan et al. [16] extended SLGDA into kernel space and proposed fast kernel SLGDA for feature extraction of HSIs. Peng et al. [17] proposed a maximum likelihood estimation-based robust joint sparse representation (JSR) model which can reduce the effect of inhomogeneous neighboring pixels or outliers. However, most of those methods focus on feature extraction in first-order vector space, which need to rearrange the high-order HSI data into vector form. Unfortunately, this kind of rearrangement often destroys the intrinsic structure of high-order data.

For the high-order data analysis, tensor algebra has been a popular and effective tool for maintaining the multidimensional structure. Up to now, it has been widely applied for high-order data completion [18], [19], denoising [20], [21], and feature extraction [25], [28], etc. Since tensor algebra can deal with the high-order data without any rearrangement, many multilinear subspace learning (i.e., tensor subspace learning) methods has been developed for feature extraction of tensor data. For example, Lu et al. [22] proposed multilinear princi-

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ple component analysis (MPCA). Zheng et al. [23] extended LPP into tensor space and proposed tensor LPP (TLPP). Considering HSI is a natural 3-D data cube, we introduced TLPP into HSI analysis and modified TLPP by integrating region covariance descriptor and manifold distance in [24] and [25], respectively. To explore the complementary information from multiple views, Zhang et al. [26] established a low-rank tensor constrained multiview subspace clustering (LT-MSC) method. To simultaneously consider sample's feature information and spatial structures, Fu et al. [27] proposed a subspace clustering method via integrating LRR and sparse coding (SC) together. Based on the low-rank and sparse constrained tensor Tucker decomposition, a patch tensor-based low-rank and sparse graph (PT-SLG) algorithm was developed for feature extraction of HSIs in [28]. By constructing the graph in a core tensor space instead of the original sample space, the orthogonality graph embedding core tensor space (OGECS) was proposed in [29]. Since tensor representation can well keep the multidimensional structure of high-order data, these tensor-based feature extraction methods have shown their promising effectiveness in different applications [30], [31], [32]. However, those tensor subspace learning methods are developed based on multiple matrix projections, which independently transform the tensor data into a low-dimensional space mode by mode and ignore the relationships among different modes.

Recently, Kilmer and Martin [33] defined a novel tensor-tensor product operator termed t-product and a new higher-order extension of singular value decomposition (SVD), named tensor SVD (t-SVD). Based on this new operator, the definition of tensor nuclear norm (TNN) was given in [34]. The t-product operation leads to matrix-like multiplication, SVD, and nuclear-norm for tensor, which have the superior advantage for processing the high-order data. Therefore, many denoising and data recovery methods were proposed and applied in various areas by using t-SVD. For example, Zhang et al. developed a novel method for multilinear data completion and denoising based on t-SVD in [35] and [36]. A novel tensor factorization-based t-product operation was proposed for low-rank tensor completion by Zhou et al. [37]. Hu et al. [38] proposed a novel low-rank estimation method for video completion by using twist tensor nuclear norm (t-TNN), which is also defined on t-SVD. Yin et al. [39] developed a multiview subspace clustering by adopting a t-linear combination constructed on tensorial t-product with sparse and low-rank constraints. Tensor robust principle component analysis (TRPCA) was proposed for exactly recovering the low-rank and sparse components of data [40]. By adding a regularization term, Xu et al. [41] constructed a Mahalanobis distance-regularized TRPCA (TRPCA-MD) model for anomaly detection.

In fact, the above works have validated the advantages of the t-product-based methods for high-order data processing, especially for noise corrupted tensor data. Among them, TRPCA is the most widely applied in practice due to the excellent robustness on noise and outliers. Regrettably, TRPCA generally decomposes data into a low-rank part and a sparse term, which have the same size as the raw data. That means TRPCA failed to solve the problems caused by dimensionality increase in high-order data processing. However, HSI data not only inevitably contains noise corruptions (especially the sparse noises, such as impulse noise, water absorption bands, strips, etc.) but also suffers from the problem of high dimensionality, which requires achieving both noise suppression and dimensionality reduction during feature extraction. To this end, this article attempts to develop a robust feature extraction method with the t-product-based tensor concepts such that data completion/denoising and dimensionality reduction can be conducted simultaneously for improving the effectiveness of HSI processing.

In detail, to overcome the drawbacks of matrix projection in high-order data processing, a t-product projection is defined for tensor data transformation based on the new tensor concepts. Inspired by the outstanding robustness of TRPCA, the tensor low-rank and sparse decomposition is considered to be used for enhancing the robustness of feature extraction. Afterward, by integrating t-product projection learning into tensor low-rank and sparse decomposition, a novel multilinear subspace learning model named t-linear tensor subspace learning (tLTSL) is constructed for robust feature extraction of HSIs. The proposed tLTSL method maintains the robustness on noise corruptions and alleviates the problem of "curse of dimensionality" simultaneously. In summary, this article contains the following key contributions.

1) To overcome the shortage of matrix projection for high-order data processing, t-product projection is defined for tensor data transformation, which the can maximally capture the intrinsic structure of tensor data. Meanwhile, a new type of multilinear subspace learning named tLTSL is developed on t-product projection for robust feature extraction of HSIs, which changes the traditional matrix projection-based subspace learning paradigm. Besides, the t-product projection can exploit the global spatial structure for HSI feature extraction.

2) The tLTSL method integrates the two separate steps (i.e., denoising and dimensionality reduction) into one model to improve the efficiency of data processing, which can prune data by the tensor low-rank and sparse decomposition, and simultaneously transform data into a low-dimensional space via the learned t-product projection. Furtherly, an important proposition related to tensor rank is proofed to interpret the meaning of tLTSL model.

3) An iterative algorithm is designed for solving the proposed tLTSL model and the computational complexity of the iterative algorithm is theoretically discussed. In final, extensive experiments are conducted to demonstrate the superiority of tLTSL by using simulated and real noise corrupted HSI data.

The rest of this article is organized as follows. Section II introduces the notations and some important definitions. Section III describes the construction and optimization of the proposed tLTSL model. In Section IV, we present the experimental results and analysis. Finally, Section V draws the conclusion.

II. Notations and Definitions

Throughout this article, tensors are denoted by Euler script letters, e.g., $\mathcal{X}$. Matrices are denoted by uppercase letters, e.g., X, vectors are denoted by boldface lowercase letters,

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e.g., $\boldsymbol{x}$, and scalars are denoted by lowercase letters, e.g., m, n. For a third-order tensor $\mathcal{X}$, its (i, j, k)-th element denoted as $x_{i,j,k}$ or $\mathcal{X}(i, j, k)$, $\mathcal{X}_{(:,i,j)}, \mathcal{X}_{(i,:,j)}$, and $\mathcal{X}_{(i,j,:)}$ represent the (i, j)-th mode-1, mode-2, and mode-3 fiber (or tube), respectively, while the i-th frontal, horizontal, and lateral slices are denoted by $\mathcal{X}_{(:,:,i)}$ or $\mathcal{X}^{(i)}$, $\mathcal{X}_{(i,:,:)}$, and $\mathcal{X}_{(:,i,:)}$, respectively. Similar to the notations in MATLAB, $\widetilde{\mathcal{X}} = \text{fft}(\mathcal{X}, [\ ], 3)$ denotes the fast Fourier transform (FFT) along mode-3 of $\mathcal{X}$. Inversely, $\mathcal{X} = \text{ifft}(\widetilde{\mathcal{X}}, [\ ], 3)$. We denote the trace of matrix with $trace(\cdot)$ and use $\|\cdot\|_F$ to represent the Frobenius norm of matrix and tensor.

For a tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its block circulant matrix is defined as

$$\text{circ}(\mathcal{X}) = \begin{bmatrix} \mathcal{X}^{(1)} & \mathcal{X}^{(n_3)} & \cdots & \mathcal{X}^{(2)} \\ \mathcal{X}^{(2)} & \mathcal{X}^{(1)} & \cdots & \mathcal{X}^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathcal{X}^{(n_3)} & \mathcal{X}^{(2)} & \cdots & \mathcal{X}^{(1)} \end{bmatrix}.$$

Two operators used for defining the tensor-tensor product are defined as

$$\text{unfold}(\mathcal{X}) = \begin{bmatrix} \mathcal{X}^{(1)} \\ \mathcal{X}^{(2)} \\ \vdots \\ \mathcal{X}^{(n_3)} \end{bmatrix}, \quad \text{fold}(\text{unfold}(\mathcal{X})) = \mathcal{X}.$$

Definition 1 (t-Product): Let $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{Y} \in \mathbb{R}^{n_2 \times r \times n_3}$. Then the t-product $\mathcal{X} * \mathcal{Y}$ is defined as

$$\mathcal{X} * \mathcal{Y} = \text{fold}(\text{circ}(\mathcal{X}) \cdot \text{unfold}(\mathcal{Y})). \tag{1}$$

If $\mathcal{Z} = \mathcal{X} * \mathcal{Y}$, then $\mathcal{Z}$ is a tensor of size $n_1 \times r \times n_3$.

With the definition of t-product, some useful definitions and properties can be naturally obtained. For $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, let $\widetilde{\mathcal{X}}$ be the tensor obtained by taking FFT of all the tubes along the third dimension of $\mathcal{X}$. Then, denote the block diagonal form of the tensor $\widetilde{\mathcal{X}}$ in the Fourier domain as $\overline{X}$, which is expressed as

$$\overline{X} = \text{blockdiag}\left(\widetilde{\mathcal{X}}\right) = \begin{bmatrix} \widetilde{\mathcal{X}}^{(1)} & & & \\ & \widetilde{\mathcal{X}}^{(2)} & & \\ & & \ddots & \\ & & & \widetilde{\mathcal{X}}^{(n_3)} \end{bmatrix}. \tag{2}$$

Remarkably, the block diagonal form $\overline{X}$ also can be rewritten as third-order tensor $\widetilde{\mathcal{X}}$ in the Fourier domain through an inverse operation.

Property 1: t-product is equivalent to the matrix multiplication in the Fourier domain, i.e., $\mathcal{Z} = \mathcal{X} * \mathcal{Y}$ and $\overline{Z} = \overline{X}\overline{Y}$ are equivalent.

Definition 2 (Identity Tensor): An identity tensor $\mathcal{I} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ is a tensor whose first frontal slice is the $n_1 \times n_1$ identity matrix and all other frontal slices are zero.

Definition 3 (Tensor Transpose): For $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its transpose $\mathcal{X}^T$ is an $n_2 \times n_1 \times n_3$ tensor obtained by transposing each frontal slices of $\mathcal{X}$ and then reversing the order of transposed frontal slices 2 through n_3.

Definition 4 (Orthogonal Tensor): If a tensor $\mathcal{Q} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ satisfies $\mathcal{Q} * \mathcal{Q}^T = \mathcal{Q}^T * \mathcal{Q} = \mathcal{I}$, $\mathcal{Q}$ is an orthogonal tensor.

Definition 5 (f-Diagonal Tensor): A tensor is f-diagonal if each frontal slice of the tensor is a diagonal matrix.

Definition 6 (t-SVD): Let $\mathcal{X}$ be an $n_1 \times n_2 \times n_3$ tensor, the t-SVD of $\mathcal{X}$ is given by $\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$, where $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal tensors. $\mathcal{S}$ denotes an f-diagonal tensor of size $n_1 \times n_2 \times n_3$, and $*$ represents the t-product.

Definition 7 (Tensor Tubal Rank): For a given tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its tubal rank is defined as the number of nonzero singular tubes of $\mathcal{S}$, where $\mathcal{S}$ is from the t-SVD as $\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$ and $rank(\mathcal{X}) = \#\{i : \mathcal{S}(i, i, :) \neq 0\}$.

Property 2: For $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $rank(\mathcal{X}^T * \mathcal{X}) = \text{rank}(\mathcal{X} * \mathcal{X}^T) = rank(\mathcal{X})$.

Definition 8 (Tensor Nuclear Norm): For a given tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its TNN is defined as

$$\|\mathcal{X}\|_{\circledast} = \sum_{i=1}^{\min(n_1, n_2)} \sum_{k=1}^{n_3} |\tilde{s}_{i,i,k}|$$

where $\tilde{s}_{i,i,k}$ is the (i, i, k)-th entry of $\widetilde{\mathcal{S}}$ and $\widetilde{\mathcal{S}} = fft(\mathcal{S}, [], 3)$.

Property 3: For $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $\widetilde{\mathcal{X}}$ denotes the FFT along mode-3 of $\mathcal{X}$, and $\overline{X}$ is the block diagonal form of $\widetilde{\mathcal{X}}$. Then, we have

(i) $\|\mathcal{X}\|_F^2 = trace((\mathcal{X}^T * \mathcal{X})_{(:,:,1)})$;
(ii) $\|\mathcal{X}\|_F = (1/(n_3)^{1/2})\|\widetilde{\mathcal{X}}\|_F = (1/(n_3)^{1/2})\|\overline{X}\|_F$;
(iii) $\|\mathcal{X}\|_{\circledast} = \sum_{k=1}^{n_3} \|\widetilde{\mathcal{X}}^{(k)}\|_* = \|\overline{X}\|_*$.

III. Proposed Method

This section first introduces a general tLTSL, and then formulates a robust tLTSL model. The model formulation and optimization are also detailedly presented. At last, the computational complexity of the optimal algorithm is theoretically analyzed.

A. Model Formulation

As is well known, HSI is appeared as 3-D data cube (i.e., third-order tensor), which contains 2-D spatial information and 1-D spectral information that can distinguish objects with tiny differences. However, due to the influences of environmental factors and sensor errors, the collected HSI data is inevitably disturbed by noise corruptions. Meanwhile, the high spectral resolution also leads to abundant redundant information. Thus, robust feature extraction becomes an urgent requirement in HSI processing. Currently, most of HSI feature extraction methods (such as PCA, LPP, LDA, etc.) are belonging to linear subspace learning, which are developed on the linear transformation matrix in vector space. Unfortunately, they often become inadequate when dealing with multidimensional tensor data such as HSI. They require rearranging the tensor data into very high-dimensional vectors, which not only destroy the intrinsic geometrical structure of tensor data but also result in a large number of parameters to be estimated.

To deal with those issues, multilinear (tensor) subspace learning is developed for the dimensionality reduction of tensor data, which can directly maps high-dimensional tensor data to low-dimensional subspace. Different from linear subspace

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learning, multilinear subspace learning performs dimensionality reduction via multilinear projection, which can avoid the destruction for data structures caused by the vectorization of tensor data. However, the current multilinear projection usually transforms high-dimensional tensor data to low-dimensional space mode by mode with multiple matrix projections, which take the information from each mode of tensor into consideration independently and fail to exploit the correlations among the modes. For overcoming the shortage of multiple matrix projections, we introduce a new multilinear projection termed t-product projection, which can comprehensively explore the information from all dimensions of tensor data in once transformation operation, whose definition is expressed as

Definition 9 (t-Product Projection): Given $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{A} \in \mathbb{R}^{r \times n_1 \times n_3}$. If $T(\mathcal{X}) = \mathcal{A} * \mathcal{X}$, $T : \mathbb{R}^{n_1 \times n_2 \times n_3} \to \mathbb{R}^{r \times n_2 \times n_3}$ is a multilinear transformation, where $\mathcal{A}$ is called t-product projection.

Based on the defined t-product projection, a new multilinear subspace learning model named tLTSL is developed for robust feature extraction of HSI. In tLTSL model, the tensor low-rank and sparse decomposition whose robustness has been verified by the applications of TRPCA model is integrated to learn a robust t-product projection for suppressing the negative influence of noise corruptions. Mathematically, the objective function of the tLTSL model is formulated as

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{E}_p,\mathcal{P}} \operatorname{rank}(\mathcal{L}) + \lambda \|\mathcal{E}_p\|_1 \\ &\text{s.t. } \mathcal{P}^T * \mathcal{X} = \mathcal{P}^T * \mathcal{L} + \mathcal{E}_p, \quad \mathcal{P}^T * \mathcal{P} = \mathcal{I} \end{aligned} \tag{3}$$

where $\mathcal{P} \in \mathbb{R}^{l \times d \times n}$ ($d \ll l$) is a projection tensor, $\mathcal{X} \in \mathbb{R}^{l \times m \times n}$ is the observation data tensor, $\mathcal{L}$ denotes the low-rank component of $\mathcal{X}$, and $\mathcal{E}_p \in \mathbb{R}^{d \times m \times n}$ is the sparse residual tensor in the low-dimensional space.

B. Model Analysis

In particular, the t-product projection enjoys similar properties to the linear transformation in linear space, since the new tensor concepts and their relationships are consistent with the ones for the matrix case. This implies that tLTSL will have similar physical significance with the linear subspace learning. Furthermore, the hyperspectral data is naturally a third-order tensor, thus using t-product projection for HSI processing can capture the global 3-D data structure and avoid the information loss caused by block segmentation in patch tensor-based methods. More importantly, tLTSL can alleviate the so-called "curse of dimensionality" by the learned t-product projection and simultaneously prune data via low-rank and sparse decomposition. In addition, for the feature extraction of 3-D HSI data, the projection purpose is to compress the features on spectral space. Therefore, the sparse noise in the spectral direction will be effectively suppressed via the subspace projection. However, the sparse noise on the 2-D spatial space will be maintained to the low-dimensional space in certain. Thus, a sparse constraint is still imposed on the residual tensor to model the sparse noise from the spatial dimension.

Remarkably, for the t-product projection, the following proposition holds.

Proposition 1: For arbitrarily given $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, if $\mathcal{A} \in \mathbb{R}^{r \times n_1 \times n_3}$ is a t-product projection that subjects to $\mathcal{A}^T * \mathcal{A} = \mathcal{I}$, then $rank(\mathcal{A} * \mathcal{X}) = rank(\mathcal{X})$ holds.

Proof: Since $\mathcal{A}^T * \mathcal{A} = \mathcal{I}$, when let $\mathcal{C} = \mathcal{A} * \mathcal{X}$, we have $rank(\mathcal{C}) = \operatorname{rank}(\mathcal{C}^T * \mathcal{C}) = \operatorname{rank}((\mathcal{A} * \mathcal{X})^T * (\mathcal{A} * \mathcal{X})) = \operatorname{rank}(\mathcal{X}^T * (\mathcal{A}^T * \mathcal{A}) * \mathcal{X}) = \operatorname{rank}(\mathcal{X}^T * \mathcal{X}) = \operatorname{rank}(\mathcal{X})$. □

According to Proposition 1, while letting $\mathcal{L}_p = \mathcal{P}^T * \mathcal{L}$, there have $\operatorname{rank}(\mathcal{L}) = \operatorname{rank}(\mathcal{P}^T * \mathcal{L}) = \operatorname{rank}(\mathcal{L}_p)$ in (3). Then, the tLTSL model is equivalent to

$$\begin{aligned} &\min_{\mathcal{L}_p,\mathcal{E}_p,\mathcal{P}} \operatorname{rank}(\mathcal{L}_p) + \lambda \|\mathcal{E}_p\|_1 \\ &\text{s.t. } \mathcal{P}^T * \mathcal{X} = \mathcal{L}_p + \mathcal{E}_p, \ \mathcal{P}^T * \mathcal{P} = \mathcal{I}. \end{aligned} \tag{4}$$

To this end, denoting $\mathcal{X}_p = \mathcal{P}^T * \mathcal{X}$ as the projected low-dimensional data of the noisy observation $\mathcal{X}$, the above minimizing problem can be considered to explore a t-product projection $\mathcal{P}$ such that $\mathcal{X}_p$ can be decomposed into a low-rank component $\mathcal{L}_p$ and a sparse component $\mathcal{E}_p$ similar to TRPCA, i.e., $\mathcal{X}_p = \mathcal{L}_p + \mathcal{E}_p$. In other words, the tLTSL model is to search a reasonable transformation such that it maps the original observation data into a low-dimensional space and its resulting low-dimensional representation enjoys a similar low-rank and sparse decomposition to that of the original space.

In addition, if $\mathcal{P}$ is fixed as an identity tensor with matched dimensions, the tLTSL model reduces to the TRPCA model.

C. Model Optimization

In order to solve the constructed tLTSL model, TNN is first employed to approximate the rank of $\mathcal{L}$ for avoiding the NP-hard problem of tensor rank minimizing. Correspondingly, the tLTSL model can be rewritten as

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{E}_p,\mathcal{P}} \|\mathcal{L}\|_{\circledast} + \lambda \|\mathcal{E}_p\|_1 \\ &\text{s.t. } \mathcal{P}^T * \mathcal{X} = \mathcal{P}^T * \mathcal{L} + \mathcal{E}_p, \ \mathcal{P}^T * \mathcal{P} = \mathcal{I} \end{aligned} \tag{5}$$

where $\|\cdot\|_{\circledast}$ denotes the TNN of a tensor.

Then, alternating direction method of multiplier (ADMM) can be applied to solve the tLTSL model. Let $\mathcal{L} = \mathcal{M}$, the optimization problem (5) becomes

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{E}_p,\mathcal{P}} \|\mathcal{M}\|_{\circledast} + \lambda \|\mathcal{E}_p\|_1 \\ &\text{s.t., } \mathcal{P}^T * \mathcal{X} = \mathcal{P}^T * \mathcal{L} + \mathcal{E}_p, \ \mathcal{P}^T * \mathcal{P} = \mathcal{I}, \ \mathcal{L} = \mathcal{M}. \end{aligned} \tag{6}$$

Thus, the corresponding augmented Lagrangian function can be formulated as

$$\begin{aligned} &\mathfrak{L}_\mu(\mathcal{L}, \mathcal{E}_p, \mathcal{P}, \mathcal{W}_1, \mathcal{W}_2) \\ &\quad = \|\mathcal{M}\|_{\circledast} + \lambda \|\mathcal{E}_p\|_1 \\ &\qquad + \langle \mathcal{W}_1, \mathcal{P}^T * \mathcal{X} - \mathcal{P}^T * \mathcal{L} - \mathcal{E}_p \rangle + \langle \mathcal{W}_2, \mathcal{L} - \mathcal{M} \rangle \\ &\qquad + \frac{\mu}{2} \left(\|\mathcal{P}^T * \mathcal{X} - \mathcal{P}^T * \mathcal{L} - \mathcal{E}_p\|_F^2 + \|\mathcal{L} - \mathcal{M}\|_F^2 \right). \end{aligned} \tag{7}$$

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After several simple derivations, we can obtain the following equivalent problem:

$$\begin{aligned}\mathfrak{L}_\mu(\mathcal{L},\mathcal{E}_p,\mathcal{P},\mathcal{W}_1,\mathcal{W}_2) &= \|\mathcal{M}\|_{\circledast}+\lambda\|\mathcal{E}_p\|_1 \\ &+\frac{\mu}{2}\left(\left\|\mathcal{P}^T*\mathcal{X}-\mathcal{P}^T*\mathcal{L}-\mathcal{E}_p+\frac{\mathcal{W}_1}{\mu}\right\|_F^2\right. \\ &\left.+\left\|\mathcal{L}-\mathcal{M}+\frac{\mathcal{W}_2}{\mu}\right\|_F^2\right). \end{aligned} \tag{8}$$

Then, the updating steps are as follows.

1) Updating $\mathcal{M}$: By fixing all other variables except $\mathcal{M}$, the solution to $\mathcal{M}$ can be found by minimizing the problem

$$\min_{\mathcal{M}}\frac{1}{\mu}\|\mathcal{M}\|_{\circledast}+\frac{1}{2}\left\|\mathcal{M}-\left(\mathcal{L}+\frac{\mathcal{W}_2}{\mu}\right)\right\|_F^2 \tag{9}$$

whose closed-form solution can be obtained by the following theorem [44].

Theorem 1: For $\tau>0$ and $\mathcal{A},\ \mathcal{B}\in\mathbb{R}^{n_1\times n_2\times n_3}$, the following optimization problem:

$$\min_{\mathcal{A}}\tau\|\mathcal{A}\|_{\circledast}+\frac{1}{2}\|\mathcal{A}-\mathcal{B}\|_F^2 \tag{10}$$

has a globally optimal solution, which is given by the tensor singular value shrinkage operator

$$\mathcal{A}=\mathfrak{D}_{n_3\tau}(\mathcal{B})=\mathcal{U}*\mathfrak{D}_{n_3\tau}(\mathcal{S})*\mathcal{V}^T \tag{11}$$

where $\mathcal{B}=\mathcal{U}*\mathcal{S}*\mathcal{V}^T$ and $\mathfrak{D}_{n_3\tau}(\mathcal{S})=\mathcal{S}*\mathcal{J}$. Herein, $\mathcal{J}$ is an $n_1\times n_2\times n_3$ f-diagonal tensor whose diagonal element in the Fourier domain is $\widetilde{\mathcal{J}}(i,i,j)=(1-(n_3\tau/\widetilde{\mathcal{S}}(i,i,j)))_+$.

From Theorem 1, the solution to problem (8) is

$$\mathcal{M}^\star=\mathfrak{D}_{\frac{u}{\mu}}\left(\mathcal{L}+\frac{\mathcal{W}_2}{\mu}\right). \tag{12}$$

2) Updating $\mathcal{L}$: By giving all other variables except $\mathcal{L}$, the subproblem for solving $\mathcal{L}$ is formulated as

$$\min_{\mathcal{L}}\left\|\mathcal{P}^T*\mathcal{X}-\mathcal{P}^T*\mathcal{L}-\mathcal{E}_p+\frac{\mathcal{W}_1}{\mu}\right\|_F^2+\left\|\mathcal{L}-\mathcal{M}+\frac{\mathcal{W}_2}{\mu}\right\|_F^2. \tag{13}$$

According to Property 1, the above optimization problem has the following equivalent formulation in the Fourier domain, that is,

$$\min_{\overline{L}}\|\overline{P}^T\overline{X}-\overline{P}^T\overline{L}-\overline{E}_p+\frac{\overline{W}_1}{\mu}\|_F^2+\|\overline{L}-\overline{M}+\frac{\overline{W}_2}{\mu}\|_F^2. \tag{14}$$

Then, the optimal $\overline{L}^\star$ can be computed by

$$\begin{aligned}\overline{L}^\star=&\left(\overline{P}\,\overline{P}^T+\overline{I}\right)^{-1} \\ &\times\left(\overline{P}\left(\overline{P}^T\overline{X}-\overline{E}_p+\frac{\overline{W}_1}{\mu}\right)+\left(\overline{M}-\frac{\overline{W}_2}{\mu}\right)\right).\end{aligned} \tag{15}$$

Thus, $\widetilde{\mathcal{L}}^\star$ can be obtained by reorganizing $\overline{L}^\star$ into a third-order tensor and $\mathcal{L}^\star=\text{ifft}(\widetilde{\mathcal{L}}^\star,[\,],3)$. The optimization procedure of solving $\mathcal{L}$ is described in Algorithm 1.

Algorithm 1 Optimization of $\mathcal{L}$

Input : Third-order tensors $\mathcal{X},\mathcal{P},\mathcal{E}_p,\mathcal{M},\mathcal{W}_1,\mathcal{W}_2,\mathcal{I}$ and the parameter μ;

1 Transform $\mathcal{X},\mathcal{P},\mathcal{E}_p,\mathcal{M},\mathcal{W}_1,\mathcal{W}_2$ into Fourier domain by $\widetilde{\mathcal{P}}=fft(\mathcal{P},[\,],3)$, $\widetilde{\mathcal{X}}=fft(\mathcal{X},[\,],3)$, $\widetilde{\mathcal{M}}=fft(\mathcal{M},[\,],3)$, $\widetilde{\mathcal{E}_p}=fft(\mathcal{E}_p,[\,],3)$, $\widetilde{\mathcal{W}_1}=fft(\mathcal{W}_1,[\,],3)$, $\widetilde{\mathcal{W}_2}=fft(\mathcal{W}_2,[\,],3)$, and $\widetilde{\mathcal{I}}=fft(\mathcal{I},[\,],3)$;

2 Compute $\overline{X},\overline{P},\overline{E}_p,\overline{M},\overline{W}_1,\overline{W}_2,\overline{I}$ by Definition 2;

3 Update $\overline{L}$ by (15);

4 Reorganize $\overline{L}$ into third-order tensor $\widetilde{\mathcal{L}}$;

5 Compute $\mathcal{L}$ by $\mathcal{L}=ifft(\widetilde{\mathcal{L}},[\,],3)$;

Output: $\mathcal{L}$.

3) Updating $\mathcal{E}_p$: By keeping other variables unchanged in (7), we have

$$\begin{aligned}\mathcal{E}_p^\star=\arg\min_{\mathcal{E}_p}\frac{\lambda}{\mu}\|\mathcal{E}_p\|_1+\frac{1}{2}\left\|\mathcal{E}_p-\left(\mathcal{P}^T*\mathcal{X}\right.\right. \\ \left.\left.-\mathcal{P}^T*\mathcal{L}+\frac{\mathcal{W}_1}{\mu}\right)\right\|_F^2.\end{aligned} \tag{16}$$

According to [40], the closed-form solution to the above problem is given by

$$\mathcal{E}_p^\star=\mathfrak{S}_{\frac{\lambda}{\mu}}\left(\mathcal{P}^T*\mathcal{X}-\mathcal{P}^T*\mathcal{L}+\frac{\mathcal{W}_1}{\mu}\right) \tag{17}$$

where $\mathfrak{S}_\tau(x)$ denotes the soft-thresholding operator, which is defined as

$$\mathfrak{S}_\tau(x)=\text{sgn}(x)\max(|x|-\tau,\ 0). \tag{18}$$

4) Updating $\mathcal{P}$: Given all other variables except $\mathcal{P}$, the minimization of (7) can be rewritten as

$$\begin{aligned}&\min_{\mathcal{P}}\frac{\mu}{2}\left\|\mathcal{P}^T*(\mathcal{X}-\mathcal{L})-\left(\mathcal{E}_p-\frac{\mathcal{W}_1}{\mu}\right)\right\|_F^2 \\ &\text{s.t. }\mathcal{P}^T*\mathcal{P}=\mathcal{I}.\end{aligned} \tag{19}$$

Similar to step 2), the above problem can be solved by using the equivalent formula in the Fourier domain, that is,

$$\begin{aligned}\overline{P}^\star&=\arg\min_{\overline{P}}\frac{\mu}{2}\left\|\overline{P}^T\overline{\mathcal{X}}-\overline{P}^T\overline{L}-\overline{E}_p+\frac{\overline{W}_1}{\mu}\right\|_F^2 \\ &=\arg\min_{\overline{P}}\frac{\mu}{2}\left\|\overline{P}^T\left(\overline{X}-\overline{L}\right)-\left(\overline{E}_p-\frac{\overline{W}_1}{\mu}\right)\right\|_F^2 \\ &\text{s.t. }\overline{P}^T\overline{P}=\overline{I}.\end{aligned} \tag{20}$$

Theorem 2 [45]: For two given matrices A and B, solving the following constrained minimization problem:

$$\begin{aligned}P^\star=\arg\min_P\left\|P^TA-B\right\|_F^2 \\ \text{s.t. }P^TP=I\end{aligned} \tag{21}$$

is equivalent to finding the nearest orthogonal matrix to a given matrix $M=AB^T$. Suppose the SVD of M is $U\Sigma V^T$. Then $P^\star=UV^T$.

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Algorithm 2 Optimization of $\mathcal{P}$

Input : Third-order tensors $\mathcal{X}, \mathcal{L}, \mathcal{E}, \mathcal{W}_1$, and the parameter μ;

Transform $\mathcal{X}, \mathcal{L}, \mathcal{E}, \mathcal{W}_1$ into Fourier domain by $\widetilde{\mathcal{X}} = fft(\mathcal{X}, [\,], 3)$, $\widetilde{\mathcal{L}} = fft(\mathcal{L}, [\,], 3)$, $\widetilde{\mathcal{E}} = fft(\mathcal{E}, [\,], 3)$, and $\widetilde{\mathcal{W}}_1 = fft(\mathcal{W}_1, [\,], 3)$;
Compute $\overline{X}, \overline{L}, \overline{E}_p$, and $\overline{W}_1$ by Definition 2;
Perform SVD on $(\overline{X} - \overline{L})(\overline{E}_p - \frac{\overline{W}_1}{\mu})^T$ to get $\overline{U}, \overline{V}$;
Update $\overline{P}$ by $\overline{P} = \overline{U}\overline{V}^T$;
Reorganize $\overline{P}$ into third-order tensor $\widetilde{\mathcal{P}}$;
Compute $\mathcal{P}$ by $\mathcal{P} = ifft(\widetilde{\mathcal{P}}, [\,], 3)$;

Output: $\mathcal{P}$.

Algorithm 3 ADMM for Solving tLTSL

Input : Third-order tensors $\mathcal{X} \in \mathbb{R}^{l\times m\times n}$ and the parameters λ, μ_0, η, and μ_{max};

Initialize: $\mathcal{L}^{[0]} = \mathcal{O}_{l\times m\times n}$, $\mathcal{E}_p^{[0]} = \mathcal{O}_{r\times m\times n}$, $\mathcal{P}^{[0]} = \mathcal{I}_{l\times r\times n}$, $\mathcal{W}_1^{[0]} = \mathcal{E}_p^{[0]}$, $\mathcal{W}_2^{[0]} = \mathcal{L}^{[0]}$;

repeat
  Updating $\mathcal{M}^{[t+1]}$ by Eq. (12);
  Computing $\mathcal{L}^{[t+1]}$ by the Algorithm 1;
  Updating $\mathcal{E}_p^{[t+1]}$ by Eq. (17);
  Calculating $\mathcal{P}^{[t+1]}$ by the Algorithm 2;
  Updating $\mathcal{W}_1^{[t+1]}$, $\mathcal{W}_2^{[t+1]}$, and $\mu^{[t+1]}$ by (23);
  $t \leftarrow t + 1$;
until *convergence*;

Output : $\mathcal{P}$, $\mathcal{L}$, $\mathcal{E}_p$.

From Theorem 2, it is easy to solve the problem (20) through SVD, that is,

$$\overline{U}\overline{S}\overline{V}^T = \mathcal{SVD}\left(\left(\overline{X} - \overline{L}\right)\left(\overline{E}_p - \frac{\overline{W}_1}{\mu}\right)^T\right). \tag{22}$$

Thus, we can obtain the optimal projection in Fourier domain as $\overline{P}^\star = \overline{U}\overline{V}^T$. Then, by operating $\overline{P}^\star$ into third-order tensor $\widetilde{\mathcal{P}}^\star$, $\mathcal{P}^\star = \text{ifft}(\widetilde{\mathcal{P}}^\star, [\,], 3)$. The procedure of optimizing $\mathcal{P}$ is completely summarized in Algorithm 2.

5) Updating Parameters $\mathcal{W}_1$, $\mathcal{W}_2$, and μ: The Lagrangian multipliers $\mathcal{W}_1$, $\mathcal{W}_2$, and parameter μ are updated as

$$\begin{aligned} \mathcal{W}_1 &= \mathcal{W}_1 + \mu\left(\mathcal{P}^T * \mathcal{X} - \mathcal{P}^T * \mathcal{L} - \mathcal{E}_p\right) \\ \mathcal{W}_2 &= \mathcal{W}_2 + \mu(\mathcal{L} - \mathcal{M}) \\ \mu &= \min(\rho\mu, \mu_{\max}). \end{aligned} \tag{23}$$

The complete optimization procedure of solving the tLTSL model is summarized in Algorithm 3.

D. Complexity Analysis

The optimization algorithm for solving the tLTSL model is iterative, whose total computational cost can be obtained with the cost of each iteration being well investigated. The time complexity of each iteration in the proposed algorithm mainly comes from updating the variables $\mathcal{M}$, $\mathcal{L}$, $\mathcal{E}_p$, and $\mathcal{P}$. In detail, the main cost for updating $\mathcal{M}$ comes from t-SVD that needs $O(lmn\log(n) + \min(l, m)lmn)$. Algorithm 1 for updating $\mathcal{L}$ involves FFT, matrix inverse, and matrix product, whose time complexities are $O(lmn\log(n))$, $O(l^3n)$, and $O(dmln)$, respectively. Thus, the total cost of updating $\mathcal{L}$ is $O(lmn\log(n) + l^3n + rmln)$. While ignoring the negligible cost of thresholding operation, the computation of solving $\mathcal{E}_p$ takes the cost of order $O(\max(d, m)ln\log(n) + dmln)$. The main cost of Algorithm 2 for solving $\mathcal{P}$ is from FFT, matrix product, and SVD, which requires a total complexity of order $O(lmn\log(n) + dmln + dl^2n)$. Assume that $n > l > d$, m and n are comparable. Then the total cost of each iteration is at most $O(ln^2\log(n) + l^3n + l^2n^2)$. If Algorithm 3 needs T iterations to convergence, the total computation complexity for solving tLTSL is $O(T(ln^2\log(n) + l^3n + l^2n^2))$.

IV. Experimental Results and Analysis

In this section, a series of experiments are conducted on the simulated and real noisy HSI data to verify the effectiveness and robustness of the proposed tLTSL algorithm. Several state-of-the-art methods including MPCA, TLPP, PT-SLG, and OGECS are utilized for comparison. The experiments are performed on a computer with a CORE i7 3.00GH and 8G memory by MATLAB.

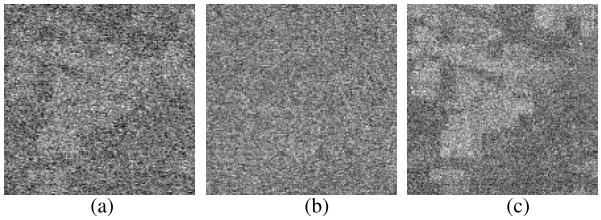
A. Data Description

Indian Pines [46] is one of the most widely used datasets, which was acquired by airborne visible/infrared imaging spectrometer (AVIRIS) sensor over an agricultural area in norther Indiana of USA in 1992. The original Indian Pines data contains 220 spectral bands from the wavelength range of 0.4–2.45 μm and covers a region of 145 $\times$ 145 pixels with 16 groundtruth classes. In the original Indian Pines data, there are 20 bands covering the region of water absorption, i.e., bands 104–108, 150–163, 220. The Indian Pines data with bands covering the region of water absorption removed is employed for verifying the proposed method. Pavia University [46] is another benchmark dataset, which was collected by the reflective optics system imaging spectrometer (ROSIS) sensor in Italy. There are 103 bands remained after removing 12 noisy bands with a spectral coverage from 0.43 to 0.86 μm. A total of 610 $\times$ 340 pixels with nine different land-cover classes are included in the image.

In our experiments, two kinds of corrupted data, i.e., simulated and real noise corrupted data, are utilized to verify the robustness of the proposed method. The Indian Pines and Pavia University data with some simulated noise similar to [20] and [47] are first employed to demonstrate the superiority of the proposed method. In detail, the following three types of noise are added to the considered datasets for simulating the real HSI noise scenarios.

1) Noise Type 1: Each band is corrupted by the noise with different intensities. Specifically, different distributions of zero-mean Gaussian noise are added to individual bands of the considered dataset. The corresponding variances are randomly selected from 0 to 0.5.

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(a) (b) (c)

Fig. 1. Some example images of the real noisy bands in the original Indian Pines data. (a) 106th band. (b) 151th band. (c) 220th band.

2) Noise Type 2: The zero-mean Gaussian noise that has the same variances in *Noise Type 1* are first added to each band. Moreover, each band is also polluted by different intensities of pepper and salt noise with ratios in [0–0.3].

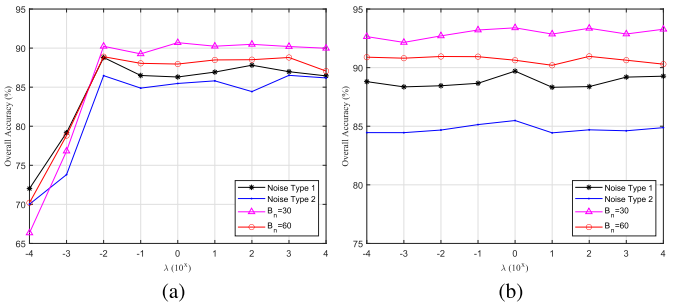
3) Noise Type 3: Considering the fact that the noise often only appears in minority bands and the noise is usually variant in different bands. B_n randomly selected bands are polluted by different distributions of zero-mean Gaussian noise with different variances randomly selected from 0 to 0.5 and simultaneously B_n bands are randomly added pepper and salt noise with ratios in [0–0.5]. In the experiments, the cases of $B_n = 30$ and $B_n = 60$ are considered, respectively.

Then, the original Indian Pines data including the bands covering the region of water absorption (which are real noisy bands) is also utilized to further evaluate the robustness of the proposed method on real noise. Some example images of real noisy bands in original Indian Pines are shown in Fig. 1. Since the original Pavia University data is unavailable, it is not included in the experiments on real noisy HSI data.

B. Experiment Setup

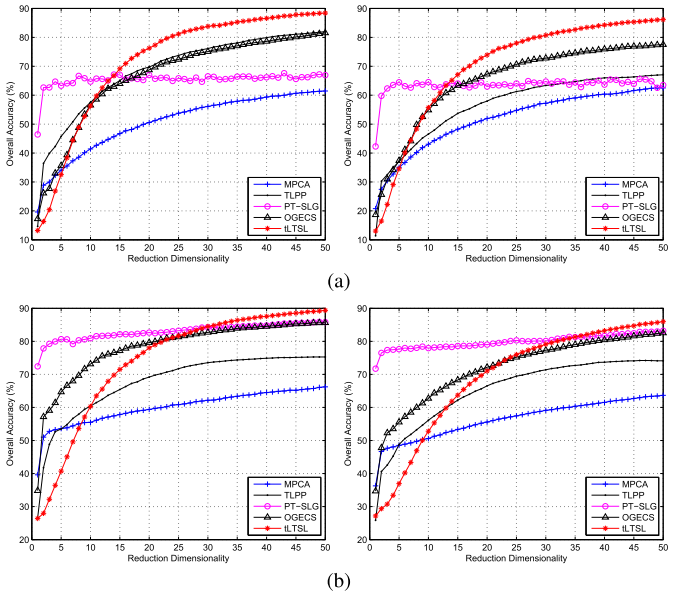
The previous work in [24], [25], [26], [27], and [28] has shown that multilinear subspace learning methods are more effective than the linear ones for HSI processing. Thus, the performance of tLTSL under the simulated and real noise corruption is evaluated and compared with some of typical matrix projection-based multilinear subspace learning methods including MPCA, TLPP, PT-SLG, and OGECS. Among these methods, MPCA is a classical unsupervised multiple matrix projections learning method. TLPP and PT-SLG are two patch tensor-based graph embedding methods. OGECS is the most recently proposed HSI dimensionality reduction method which jointly considers both spectral redundancy and spatial continuity through multiple matrix projections with graph embedding in core tensor space. In our experiments, 3% of samples from each class are randomly selected for training and the rest are used as testing samples. The classification accuracy such as overall accuracy (OA), per-class accuracy, average accuracy (AA), and Kappa Coefficient (κ) are used to quantitatively evaluate the extracted features. All classification experiments are conducted by the k nearest neighbors (kNN) algorithm and independently run ten times to eliminate the influence of sample bias. The averaged classification accuracy will be reported in the next part.

The regularization parameter λ in the tLTSL model is adjusted based on grid search, which is determined from



(a) (b)

Fig. 2. Overall classification accuracy versus different values of λ in tLTSL for HSI data with different types of noise corruptions. (a) Indian Pines. (b) Pavia University.



(a)

(b)

Fig. 3. Changes of classification accuracy according to the reduction dimensionality obtained by different feature extraction methods for the Indian Pines and Pavia University datasets corrupted by different types of noises. For each row, the figures from left to right are the experimental results of dataset with corruptions of *Noise Type 1* and *Noise Type 2*, respectively. (a) Indian Pines. (b) Pavia University.

{0.0001, 0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000}. Fig. 2 shows the experimental results of parameter tuning for both HSI datasets with different types of noise corruptions. It can be seen that setting the values of λ as 0.01 and 1 are acceptable for Indian Pines and Pavia University, respectively. The parameters of the compared methods for the following experiments are manually optimized according to their default strategies under all scenarios.

C. Experimental Results of Simulated Noisy HSI Data

The experimental results of HSI data with the corruptions of *Noise Type 1* and *Noise Type 2* are presented in Tables I and II, and Figs. 3–5. Fig. 3 shows the changes of classification accuracy versus the reduction dimensionality obtained by different feature extraction methods for the Indian Pines and Pavia University datasets corrupted by different types of noises. It is easy to find that the classification accuracy of all involved feature extraction methods is increasing sharply when the

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TABLE I

CLASSIFICATION ACCURACY(%) AND STANDARD DEVIATION FOR THE INDIAN PINES DATA CORRUPTED BY *Noise Type 1* AND *Noise Type 2*

	Nosie Type 1					*Nosie Type 2*				
Class #	MPCA	TLPP	PT-SLG	OGECS	tLTSL	MPCA	TLPP	PT-SLG	OGECS	tLTSL
1	89.35 ±10.32	99.07 ±2.07	65.15 ±8.21	99.07 ±2.07	**99.54 ±1.04**	96.76 ±7.25	82.87 ±17.07	59.09 ±11.90	99.54 ±1.04	**100.00 ±0.00**
2	59.64 ±3.82	**89.27 ±2.07**	66.70 ±7.86	77.67 ±3.93	84.52 ±2.80	60.99 ±10.85	72.65 ±12.74	62.79 ±5.24	74.51 ±4.58	**85.58 ±3.76**
3	53.69 ±5.71	70.91 ±6.54	56.98 ±6.00	65.61 ±5.58	**82.17 ±7.14**	51.49 ±8.40	56.34 ±15.40	53.69 ±2.17	59.05 ±9.04	**79.67 ±4.06**
4	62.41 ±4.22	81.86 ±5.73	56.62 ±4.31	74.45 ±8.18	**92.14 ±1.96**	70.41 ±12.02	71.00 ±12.96	66.08 ±9.87	68.21 ±16.23	**87.96 ±6.93**
5	53.38 ±6.43	77.26 ±6.80	64.85 ±5.65	73.17 ±5.90	**85.00 ±6.44**	52.24 ±10.35	60.95 ±7.77	60.29 ±7.61	70.61 ±2.63	**80.31 ±6.24**
6	50.78 ±6.14	83.15 ±2.66	66.55 ±6.15	84.53 ±3.18	**86.37 ±2.40**	56.59 ±12.64	68.22 ±10.77	56.36 ±6.54	81.90 ±2.02	**86.37 ±4.91**
7	98.15 ±2.62	**100.00 ±0.00**	72.84 ±7.61	99.07 ±2.07	**100.00 ±0.00**	99.07 ±2.07	82.41 ±6.91	62.96 ±11.72	**100.00 ±0.00**	98.15 ±4.14
8	57.76 ±5.80	87.68 ±4.78	70.05 ±6.42	80.78 ±7.67	**92.06 ±2.98**	60.31 ±13.74	71.52 ±12.42	60.33 ±5.32	83.08 ±7.23	**89.37 ±4.41**
9	93.33 ±7.45	**100.00 ±0.00**	47.37 ±3.21	**100.00 ±0.00**	98.33 ±3.73	100.00 ±0.00	83.33 ±7.27	40.35 ±11.64	96.67 ±7.45	**100.00 ±0.00**
10	50.94 ±1.94	73.38 ±3.60	57.36 ±6.03	70.57 ±2.76	**83.97 ±3.02**	52.33 ±11.24	58.87 ±6.70	56.12 ±3.95	64.16 ±10.33	**81.02 ±5.35**
11	66.15 ±4.39	79.33 ±3.82	74.49 ±4.49	90.14 ±1.38	**92.27 ±0.38**	67.68 ±7.74	65.87 ±9.53	73.41 ±2.63	86.36 ±9.18	**89.23 ±1.96**
12	49.16 ±3.40	69.65 ±4.86	54.20 ±8.86	67.42 ±3.50	**76.03 ±5.61**	49.77 ±7.44	58.52 ±6.42	56.99 ±6.06	60.52 ±13.60	**76.43 ±3.23**
13	70.43 ±7.55	92.65 ±3.23	60.44 ±10.86	90.26 ±4.22	**94.53 ±4.27**	68.03 ±12.54	71.37 ±12.25	57.07 ±8.75	**88.38 ±6.04**	88.03 ±7.73
14	80.71 ±5.03	92.41 ±1.08	80.60 ±7.71	**96.77 ±0.83**	96.05 ±1.59	80.86 ±8.03	78.40 ±5.08	73.31 ±6.42	**94.55 ±2.30**	94.34 ±3.58
15	56.24 ±5.39	87.48 ±6.18	54.50 ±7.77	74.06 ±6.77	**91.22 ±3.26**	52.27 ±11.83	63.19 ±8.84	51.87 ±5.64	61.19 ±9.57	**79.23 ±9.04**
16	85.74 ±11.20	96.39 ±3.41	47.41 ±13.39	**97.19 ±4.27**	96.59 ±2.99	87.35 ±8.54	82.33 ±6.83	39.26 ±11.34	98.39 ±1.66	**99.80 ±0.45**
OA	61.46 ±2.02	81.99 ±1.75	66.93 ±2.73	81.55 ±1.17	**88.41 ±1.37**	62.54 ±8.99	67.07 ±6.02	63.07 ±0.53	77.52 ±5.78	**86.15 ±2.31**
AA	67.37 ±2.69	86.28 ±1.20	62.26 ±1.94	83.80 ±1.05	**90.68 ±1.10**	69.14 ±7.70	70.49 ±5.51	58.12 ±3.21	80.44 ±4.62	**88.47 ±1.92**
κ	56.23 ±2.13	79.53 ±1.99	66.75 ±2.75	78.73 ±1.38	**88.37 ±1.38**	57.33 ±10.20	64.84 ±9.03	63.26 ±0.54	74.11 ±6.49	**86.10 ±2.33**

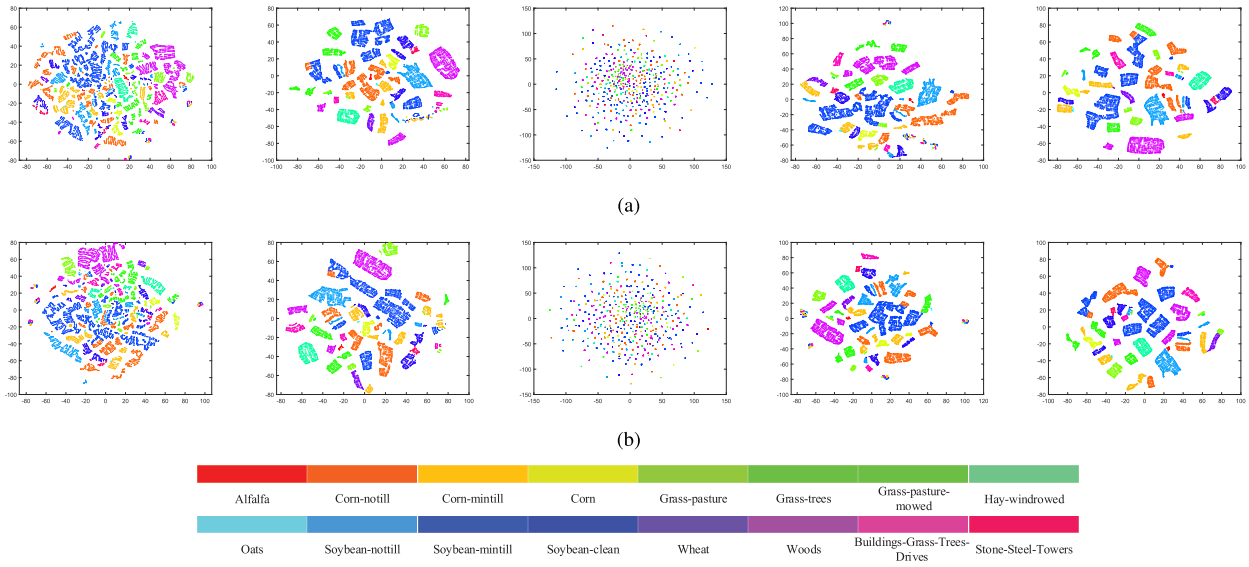


Fig. 4. Visualization of different classes on the Indian Pines dataset with the corruptions of *Noise Type 1* and *Noise Type 2* via different feature extraction methods, respectively. For each row, the visualization maps from left to right are MPCA, TLPP, PT-SLG, OGECS, and tLTSL, respectively. (a) Indian Pines with *Noise Type 1*. (b) Indian Pines with *Noise Type 2*.

feature dimensions are less than 30. The growth of classification accuracy slowed down when the feature dimensions are larger than 30 and tended to be stable around 50 dimensions. In particular, the proposed tLTSL method yields higher

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TABLE II

CLASSIFICATION ACCURACY(%) AND STANDARD DEVIATION FOR THE PAVIA UNVERSITY DATA WITH CORRUPTIONS OF *Noise Type 1* AND *Noise Type 2*

	Nosie Type 1					*Nosie Type 2*				
Class #	MPCA	TLPP	PT-SLG	OGECS	tLTSL	MPCA	TLPP	PT-SLG	OGECS	tLTSL
1	56.07 ±1.20	61.73 ±1.28	81.01 ±4.04	81.82 ±1.44	**83.92 ±1.60**	50.54 ±3.82	65.61 ±1.38	74.66 ±2.51	73.86 ±1.39	**79.49 ±1.90**
2	84.31 ±0.72	88.15 ±0.64	95.94 ±2.17	96.22 ±0.33	**98.14 ±0.49**	82.39 ±1.38	84.19 ±0.85	95.25 ±0.37	95.37 ±0.90	**96.69 ±0.28**
3	44.60 ±1.65	64.33 ±2.91	69.64 ±6.92	71.71 ±1.96	**85.97 ±2.80**	45.93 ±1.61	64.23 ±2.09	70.40 ±1.85	69.18 ±1.99	**79.62 ±4.23**
4	46.49 ±1.71	52.20 ±0.36	56.19 ±1.31	53.11 ±1.92	**58.34 ±2.31**	35.64 ±2.95	52.83 ±1.82	54.27 ±2.59	47.15 ±1.42	**54.61 ±0.84**
5	79.67 ±3.48	92.80 ±1.67	94.21 ±2.27	**99.71 ±0.26**	89.57 ±4.80	73.49 ±2.43	89.99 ±1.84	95.78 ±1.05	**99.27 ±0.50**	83.37 ±2.41
6	59.91 ±3.53	81.99 ±1.99	93.53 ±3.05	86.22 ±1.09	**96.75 ±0.39**	57.62 ±3.25	80.49 ±1.75	88.99 ±2.72	85.80 ±3.20	**93.95 ±1.78**
7	50.76 ±1.20	67.83 ±4.16	77.09 ±7.67	75.62 ±2.60	**90.96 ±2.73**	51.11 ±4.89	75.32 ±2.92	75.94 ±5.63	77.04 ±4.87	**88.59 ±2.55**
8	47.97 ±3.00	53.58 ±0.27	73.64 ±2.72	71.29 ±2.33	**79.53 ±2.43**	44.04 ±2.45	54.13 ±1.57	69.68 ±4.07	64.85 ±3.17	**71.41 ±2.56**
9	33.04 ±4.36	49.38 ±5.30	59.77 ±5.04	**86.80 ±1.96**	59.94 ±5.82	29.16 ±3.21	45.68 ±0.74	41.68 ±6.11	**69.64 ±2.68**	49.76 ±2.73
OA	66.24 ±0.90	75.27 ±0.43	85.84 ±2.20	85.64 ±0.48	**89.38 ±0.55**	63.67 ±1.53	74.12 ±0.72	83.20 ±0.54	82.53 ±0.69	**85.96 ±0.57**
AA	54.76 ±1.40	68.00 ±1.00	77.89 ±2.49	80.28 ±0.64	**82.57 ±0.72**	52.21 ±1.93	68.05 ±0.62	74.07 ±0.76	75.80 ±0.72	**77.50 ±0.67**
κ	54.42 ±1.37	67.72 ±0.58	85.71 ±2.25	80.63 ±0.67	**89.30 ±0.56**	50.87 ±2.23	66.58 ±0.86	82.99 ±0.57	76.06 ±0.97	**85.82 ±0.59**

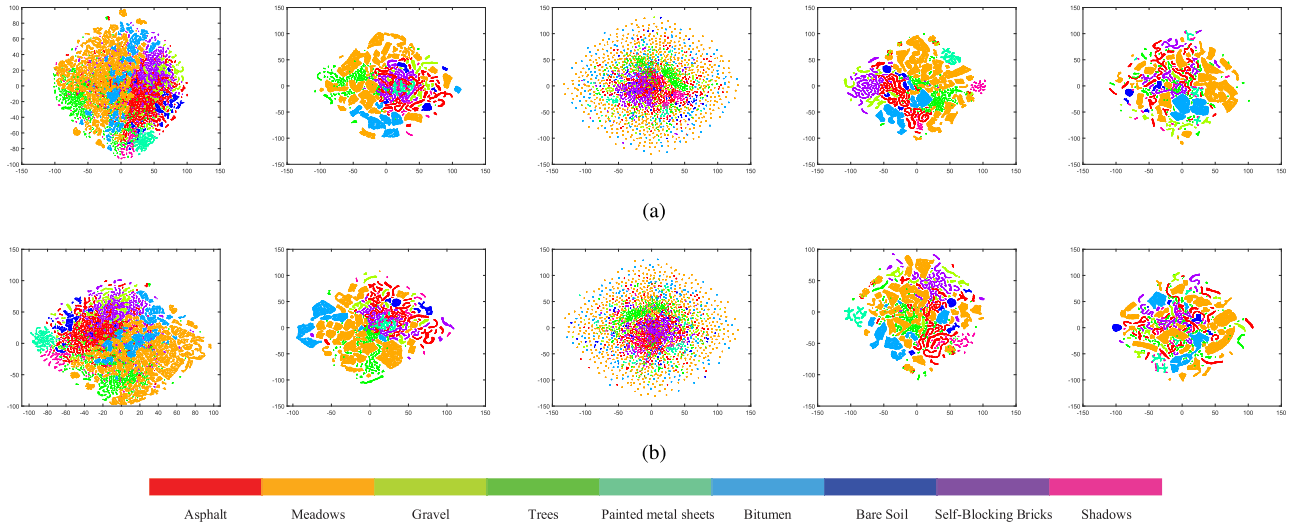


Fig. 5. Visualization of different classes on the Pavia University dataset with the corruptions of *Noise Type 1* and *Noise Type 2* via different feature extraction methods, respectively. For each row, the visualization maps from left to right are MPCA, TLPP, PT-SLG, OGECS, and tLTSL, respectively. (a) Pavia University with *Noise Type 1*. (b) Pavia University with *Noise Type 2*.

classification accuracy than other feature extraction methods for both datasets with different types of corruptions while the feature dimensions are larger than 30. To the end, the feature dimensions are set as 50 for the experiments of Indian Pines and Pavia University datasets with corruptions of *Noise Type 1* and *Noise Type 2*.

To show the classification performance more precisely, we list the per-class accuracy, OA, AA, and κ of tLTSL and the comparison methods in Table I for the Indian Pines dataset with corruptions of *Noise Type 1* and *Noise Type 2*. From Table I, it can be seen that tLTSL not only provides the best classification performance in OA, AA, and κ, but also achieves higher per-class accuracy than the comparison methods for most classes. In particular, tLTSL obtains increases in OA, AA, and κ of more than 6.4%, 4.4%, and 7.8% for *Noise type 1*. For *Noise type 2*, tLTSL outperforms the other comparison methods with increases of over 8.6%, 8%, and 11.9% in OA, AA, and κ, respectively. Table II lists the similar results for Pavia University with corruptions of *Noise Type 1* and *Noise Type 2*. The proposed tLTSL method achieves the highest OA,

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TABLE III

CLASSIFICATION ACCURACY(%) AND STANDARD DEVIATION FOR THE TWO DATASETS WITH THE CORRUPTION OF *Noise Type 3*

Dataset		Indian Pines					Pavia University				
Methods		MPCA	TLPP	PT-SLG	OGECS	tLTSL	MPCA	TLPP	PT-SLG	OGECS	tLTSL
$B_n = 30$	OA	65.01 ±2.44	87.32 ±0.97	67.88 ±1.26	83.87 ±0.76	**89.95 ±0.97**	84.93 ±1.93	87.51 ±2.45	87.36 ±0.66	89.63 ±0.28	**93.38 ±0.33**
	AA	71.66 ±2.19	90.65 ±0.48	63.88 ±2.03	87.55 ±0.76	**92.58 ±0.77**	79.40 ±2.01	83.63 ±2.39	81.03 ±1.49	83.58 ±0.44	**87.72 ±0.70**
	κ	60.28 ±2.77	85.57 ±1.09	67.71 ±1.30	81.63 ±0.87	**89.92 ±0.98**	79.91 ±2.52	83.44 ±2.58	87.26 ±0.66	86.16 ±0.37	**93.35 ±0.34**
$B_n = 60$	OA	62.75 ±1.49	87.58 ±1.26	67.12 ±2.19	83.64 ±0.64	**88.85 ±0.84**	72.53 ±1.15	77.42 ±1.05	87.01 ±0.30	86.50 ±0.56	**90.09 ±0.68**
	AA	69.01 ±1.43	89.83 ±0.89	65.64 ±1.47	85.53 ±0.74	**90.84 ±1.25**	62.92 ±1.10	70.70 ±1.02	79.84 ±0.68	81.09 ±0.73	**83.06 ±0.76**
	κ	57.55 ±1.65	85.86 ±1.42	66.93 ±2.21	81.18 ±0.73	**88.82 ±0.85**	63.16 ±1.44	70.43 ±1.32	86.90 ±0.31	81.87 ±0.76	**90.02 ±0.69**

AA, and κ simultaneously, which outperforms that of the comparative methods with the increases of more than 3.5%, 2.2%, and 3.5% for *Noise Type 1* and over 2.7%, 1.7%, and 2.8% for *Noise Type 2*, respectively. Correspondingly, Figs. 4 and 5 give the visualization of feature distribution obtained by different feature extraction methods for both datasets with corruptions of *Noise Type 1* and *Noise Type 2*, which show the distributions of different classes more directly. From Figs. 4 and 5, it can be clearly seen that the feature distributions obtained by tLTSL contain more compact intraclass distribution and more obvious interclass difference than that of the comparison methods for both datasets with different types of noise corruptions. Specifically, for some classes that are highly correlated with each other, the proposed method also shows obvious interclass difference even if the data are seriously corrupted by noise. For example, the class 10 (Soybean-notill), class 11 (Soybean-mintill), and class 12 (Soybean-clean) in Indian Pines dataset, and the class 1 (Asphalt), class 6 (Bare Soil), and class 8 (Self-blocking bricks) in the Pavia University dataset.

The experimental results of Indian Pines and Pavia University datasets with the corruption of *Noise Type 3* are presented in Table III and Figs. 6–10. Fig. 6 shows the changes of classification accuracy versus the reduction dimensionality obtained by different feature extraction methods for the Indian Pines and Pavia University datasets with the corruption of *Noise Type 3*. From the curves of classification accuracy against the reduction dimensionality, we can see that the proposed tLTSL method performs better classification accuracy than the other methods, while feature dimensions are larger than reasonable values for the cases of $B_n = 30$ and $B_n = 60$, respectively. Specifically, the classification accuracy also tends to be stable at feature dimensions around 50. Therefore, we set the feature dimensions as 50 in followed experiments.

Table III lists the classification results (including the per-class accuracy, OA, AA, and κ) for both datasets when $B_n = 30$ and $B_n = 60$, respectively. Obviously, for Indian Pines with the corruption of *Noise Type 3*, the best performance of OA, as well as AA and κ, is achieved by the proposed tLTSL method, which gains the increases of more than 2.6% in OA, 1.9% in AA, 4.3% in κ for the case of $B_n = 30$, 1.2% in OA, 1% in AA, 2.9% in κ for the case of $B_n = 60$, while comparing with the other comparison methods. For the Pavia University dataset with corruption of

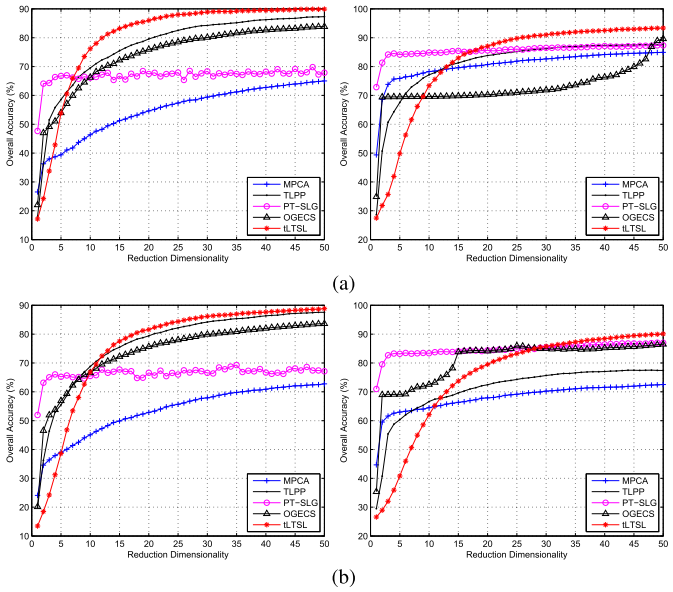


Fig. 6. Changes of classification accuracy according to the reduction dimensionality obtained by using different feature extraction methods for the HSI data with the corruptions of *Noise Type 3*. For each row, the figures from left to right are the Indian Pines and Pavia University datasets, respectively. (a) $B_n = 30$. (b) $B_n = 60$.

Noise Type 3, tLTSL also provides the highest OA, AA, and κ for both cases. Figs. 7 and 8 visually present the within-class and between-classes distributions obtained by different feature extraction methods for both datasets under different numbers of bands with noise corruption. It can be clearly seen that within-class distribution obtained by tLTSL is more compact than other methods for both datasets with corruption of *Noise Type 3*. More importantly, the differences of between-classes in the feature distribution maps obtained by tLTSL are more significant. Figs. 9 and 10 give the classification maps of all considered feature extraction methods for the Indian Pines and Pavia University datasets with corruptions of *Noise Type 3*, which show the classification results more directly. From Figs. 9 and 10, it can be found that the classification maps obtained by tLTSL contain smoother classification regions than that of the comparison methods for both datasets with different numbers of noise corruptions, which is consistent with the classification accuracy listed in Table III. More importantly,

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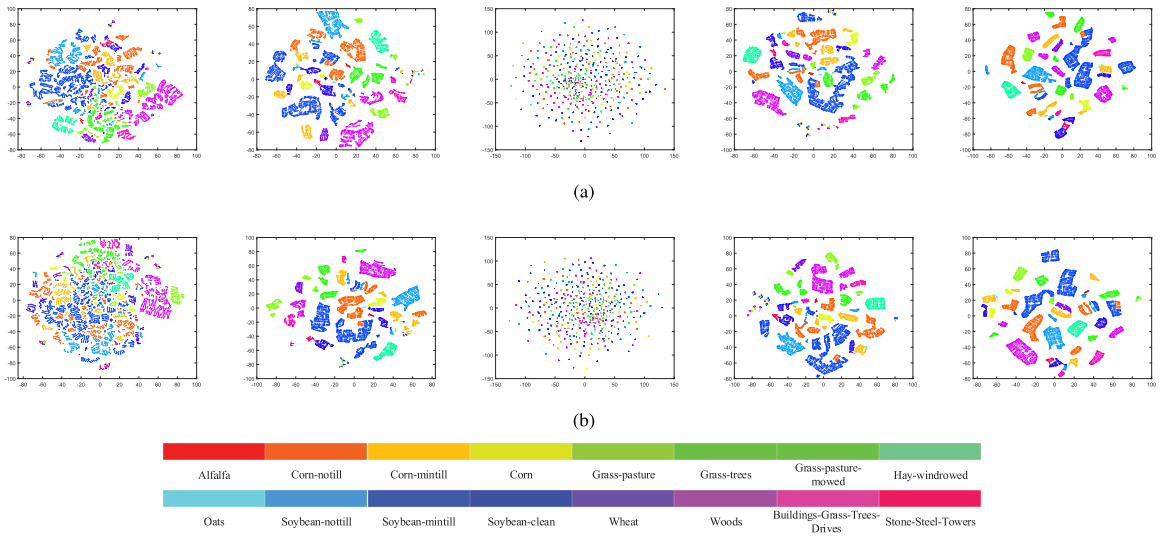


Fig. 7. Visualization of different classes on the Indian Pines dataset with the corruptions of *Noise Type 3* via different feature extraction methods, respectively. For each row, the visualization maps from left to right are MPCA, TLPP, PT-SLG, OGECS, and tLTSL, respectively. (a) $B_n = 30$ for Indian Pines. (b) $B_n = 60$ for Indian Pines.

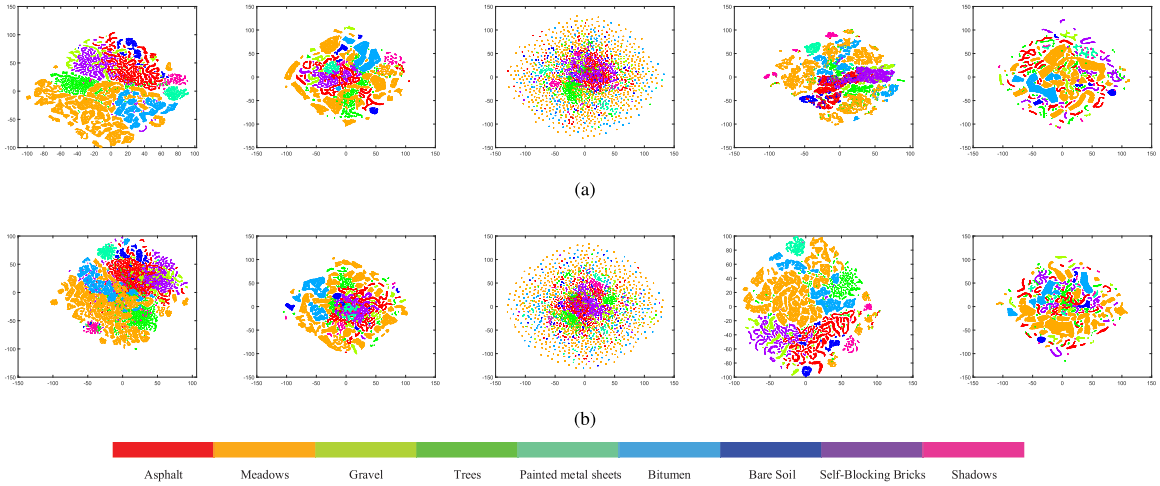


Fig. 8. Visualization of different classes on the Pavia University dataset with the corruptions of *Noise Type 3* via different feature extraction methods, respectively. For each row, the visualization maps from left to right are MPCA, TLPP, PT-SLG, OGECS, and tLTSL, respectively. (a) $B_n = 30$ for Pavia University. (b) $B_n = 60$ for Pavia University.

the classification maps gained by tLTSL contain smoother regions that is closer to the map of groundtruth.

In summary, the experimental results verify that the proposed tLTSL method performs better classification performance than other considered methods when the feature dimensions are larger than reasonable values for the HSI data with three different categories of simulated noise. The reason of the lower classification accuracy with few feature dimensions for tLTSL may be that the global spatial information plays a negative effect when the spectral dimension is relatively small. More importantly, for the situations of all bands and minority bands with different intensities of noises, the highest OA, AA, and κ are all yielded by tLTSL when the feature dimension is set as 50. In particular, for tLTSL, the gains of OA, AA, and κ on the case of all bands with noise corruptions are bigger than that on the case of minority bands with different intensities of noises. It implies that the superiority of tLTSL becomes more obvious with the intensities of noise increasing. These results demonstrate that tLTSL is more robust to noise than the comparison methods MPCA, TLPP, PT-SLG, and OGECS.

D. Experimental Results of Real Noisy HSI Data

To further demonstrate the effectiveness and robustness of the proposed method, experiments are conducted on the original Indian Pines data that contains the bands covering the region of water absorption, which are real HSI noise. Some example images from those bands are shown in Fig. 3. The associated experimental results are presented in Table IV and Fig. 11. According to Table IV, it can be found that the superiority in average OA, AA, and κ of original Indian Pines data achieved by tLTSL are prominent while comparing with other feature extraction methods such as MPCA, TLPP, PT-SLG, and OGECS. Specifically, the tLTSL method

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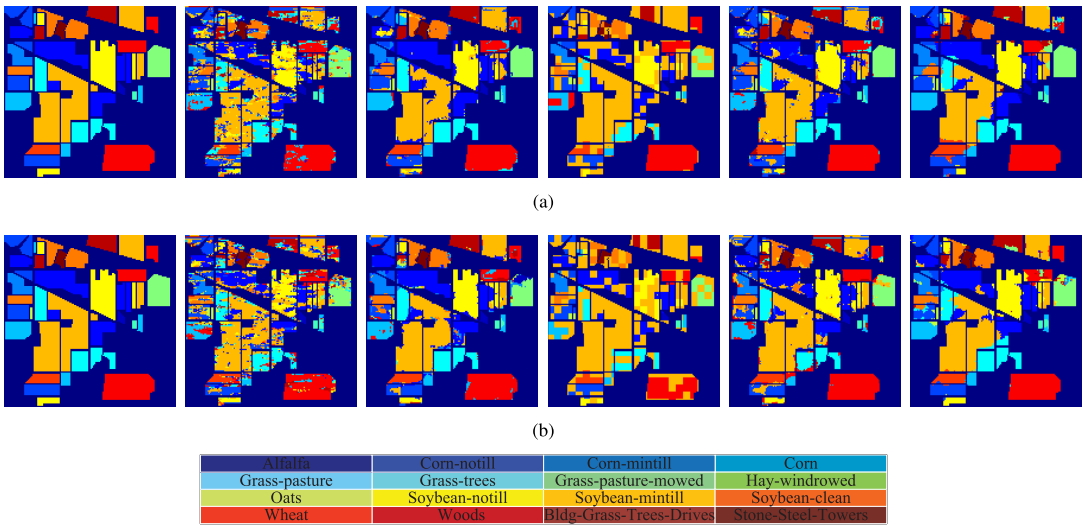


Fig. 9. Classification maps for the Indian Pines dataset with corruptions of *Noise Type 3* via different feature extraction methods. For each row, the classification maps from left to right are groundtruth, MPCA, TLPP, PT-SLG, OGECS, and tLTSL, respectively. (a) $B_n = 30$ for Indian Pines. (b) $B_n = 60$ for Indian Pines.

TABLE IV

CLASSIFICATION ACCURACY(%) AND STANDARD DEVIATION FOR ORIGINAL INDIAN PINES

#	MPCA	TLPP	PT-SLG	OGECS	tLTSL
1	90.56 ±10.63	98.61 ±2.24	46.82 ±8.66	89.72 ±6.46	**99.72 ±0.83**
2	57.23 ±2.67	79.14 ±2.82	61.62 ±3.72	62.74 ±3.02	**84.12 ±3.92**
3	49.83 ±3.24	69.83 ±1.99	62.56 ±5.11	52.29 ±4.35	**73.45 ±2.61**
4	49.52 ±5.66	77.58 ±4.82	52.36 ±11.70	52.29 ±6.77	**92.11 ±3.74**
5	62.03 ±5.93	76.29 ±6.46	**84.96 ±5.87**	63.16 ±6.26	79.30 ±6.06
6	85.14 ±3.11	**96.10 ±1.86**	88.76 ±7.93	90.72 ±2.66	87.74 ±4.30
7	96.67 ±4.44	99.44 ±1.67	**100.00 ±0.00**	97.78 ±4.44	99.44 ±1.67
8	84.81 ±5.91	**98.97 ±1.16**	96.13 ±3.51	87.89 ±4.27	90.50 ±6.32
9	89.00 ±2.21	**99.00 ±3.00**	53.68 ±12.63	94.00 ±9.17	**99.00 ±3.00**
10	66.53 ±2.87	78.93 ±2.17	74.46 ±4.00	68.39 ±3.22	**85.33 ±3.96**
11	72.97 ±2.28	85.63 ±2.77	79.15 ±3.31	75.86 ±1.46	**91.66 ±2.03**
12	36.97 ±4.08	66.70 ±4.08	53.68 ±5.62	42.45 ±4.47	**75.70 ±6.74**
13	81.23 ±7.62	**98.00 ±2.14**	85.56 ±5.60	94.15 ±5.82	94.62 ±4.35
14	91.92 ±2.07	95.91 ±1.94	93.73 ±2.32	92.86 ±3.27	**96.29 ±1.73**
15	54.60 ±8.76	70.86 ±12.46	65.78 ±9.72	61.18 ±9.68	**78.40 ±9.47**
16	93.01 ±5.70	**94.94 ±4.30**	79.89 ±6.79	94.46 ±5.17	90.96 ±7.68
OA	68.66 ±1.03	83.57 ±1.02	75.84 ±0.84	72.12 ±0.79	**86.86 ±0.80**
AA	72.63 ±1.86	86.62 ±1.18	73.66 ±2.26	76.25 ±1.07	**88.65 ±0.83**
κ	64.34 ±1.17	81.25 ±1.16	75.74 ±0.85	68.22 ±0.90	**86.82 ±0.81**

increases the performance of average OA, AA, and κ about 3.2%, 2%, and 5.5%, respectively, than some state-of-art

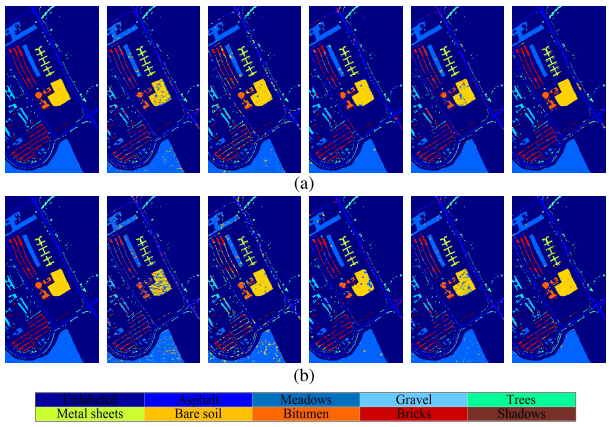


Fig. 10. Classification maps for the Pavia University dataset with corruptions of *Noise Type 3* via different feature extraction methods. For each row, the classification maps from left to right are groundtruth, MPCA, TLPP, PT-SLG, OGECS, and tLTSL, respectively. (a) $B_n = 30$ for Pavia University. (b) $B_n = 60$ for Pavia University.

methods. Meanwhile, the per-class accuracy of most classes obtained by tLTSL is better than those of compared methods. For some highly correlated classes such as class 10, class 11, and class 12, tLTSL still provides satisfactory classification accuracy for them. Furthermore, Fig. 11 shows the classification maps of all considered methods for the original Indian Pines. From those maps, it can be seen that tLTSL provides much smoother classification regions, especially in the regions of class 10, class 11, and class 12, which are more consistent to the map of groundtruth.

Overall, from the experiments on simulated and real noisy HSI data, the proposed tLTSL method outperforms the comparative methods for two different kinds of noisy HSI data. In particular, comparing with the multiple projection matrices learning methods (i.e., MPCA and TLPP), tLTSL achieves more than 3% increases of OA for the original Indian Pines. These results illustrate that tensor projection can comprehensively capture the global structure for

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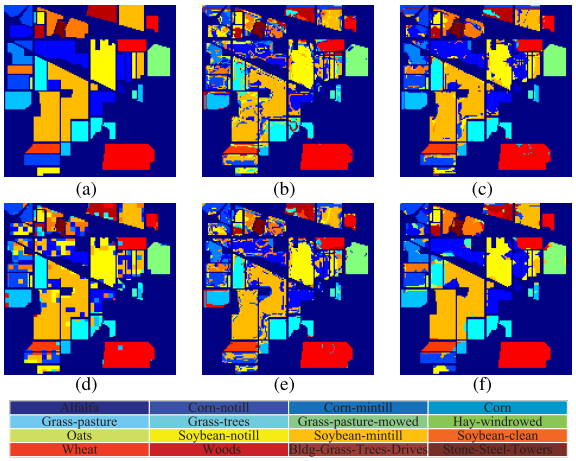


Fig. 11. Classification maps of all considered methods for original Indian Pines. (a) Groundtruth. (b) MPCA. (c) TLPP. (d) PT-SLG. (e) OGECS. (f) tLTSL.

TABLE V
CLASSIFICATION ACCURACY(%) AND STANDARD DEVIATION ON ORIGINAL INDIAN PINES OBTAINED BY TLTSL-2D, TRPCA, AND TLTSL

Methods	OA	AA	κ
tLTSL-2D	69.47 ±0.92	71.16 ±0.99	64.90 ±1.06
TRPCA	77.23 ±1.03	80.27 ±1.29	73.99 ±1.17
tLTSL	**86.86 ±0.80**	**88.65 ±0.83**	**86.82 ±0.81**

feature extraction, which is more robust than multiple matrix projections.

E. Ablation Study

As discussed in Section III-B, the proposed tLTSL model will degraded into the TRPCA model while fixing the projection $\mathcal{P}$ as an identity tensor. Thus, to verify the usefulness of the integrated t-product projection, this part will conduct some comparison experiments between TRPCA and tLTSL with the real noisy HSI data (i.e., original Indian Pines). Furthermore, the 2-D form of tLTSL denoted as tLTSL-2D will be considered to validate the usefulness of 3-D data structure.

Table V lists the classification accuracy on the original Indian Pines dataset obtained by tLTSL-D, TRPCA, and tLTSL. It is easy to see that the classification performance of tLTSL and TRPCA outperform that of tLTSL-2D. It validates that the tensor-based methods can capture the inner 3-D data structure to reduce the impacts of noise. The tLTSL method yields increases of over 9.6% in OA, 8.4% in AA, and 12.8% in κ while comparing with TRPCA. This implies that the use of t-product projection is helpful to extract more representative features of data and enhance the robustness of the method.

F. Convergence Study

In the model optimization part, the ADMM algorithm is employed for solving the proposed tLTSL model. Although the convergence of ADMM algorithm has been theoretically analyzed in many literatures, how about its convergence in real application is also important. Therefore, the convergence property of the proposed algorithm on HSI data with different noise corruptions is studied in this part.

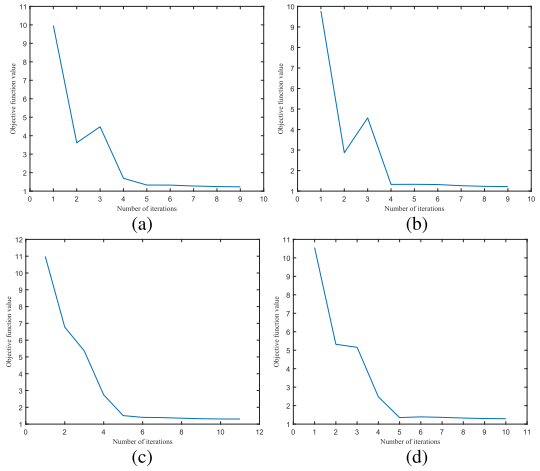


Fig. 12. Convergence property of the proposed algorithm on the Indian Pines dataset with different types of noise corruptions. (a) *Noise Type 1*. (b) *Noise Type 2*. (c) *Noise Type 3* ($B_n = 30$). (d) *Noise Type 3* ($B_n = 60$).

Fig. 12 shows the variation of the tLTSL model's objective function value with the number of iterations for the Indian Pines dataset with different types of noise corruptions. The curves in Fig. 12 illustrate the convergence property of the proposed algorithm on different kinds of noisy HSI data. It can be seen that the proposed algorithm will converge to local optimal solution with about ten loops for the Indian Pines dataset with different types of noise corruptions.

V. CONCLUSION

In this article, a novel tLTSL method is proposed for robust feature extraction of HSIs by introducing tensor projection learning into the low-rank and sparse decomposition model based on t-product operation. The low-rank and sparse decomposition is used to recover the data and remove noise corruption, which is integrated with projection learning to extract low-dimensional features. The 3-D tensor projection learned by tLTSL captures the global structure to further suppress the impact of noise for improving the quality of extracted features. Extensive experiments on HSI data with three different types of noise demonstrate the excellent performance of the proposed tLTSL method, especially the robustness against noise. The experimental results validate that tLTSL is more effective than the comparative methods. In the further work, we will make efforts to improve the tLTSL methods from two aspects. On the one hand, the tLTSL model will be extended for effectively processing large-scale HSI data. On the other hand, the prior information will be considered to improve the discriminability of extracted features.

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